

ENPH 453 Advanced Physics Laboratory
E3 Helicons in Indium

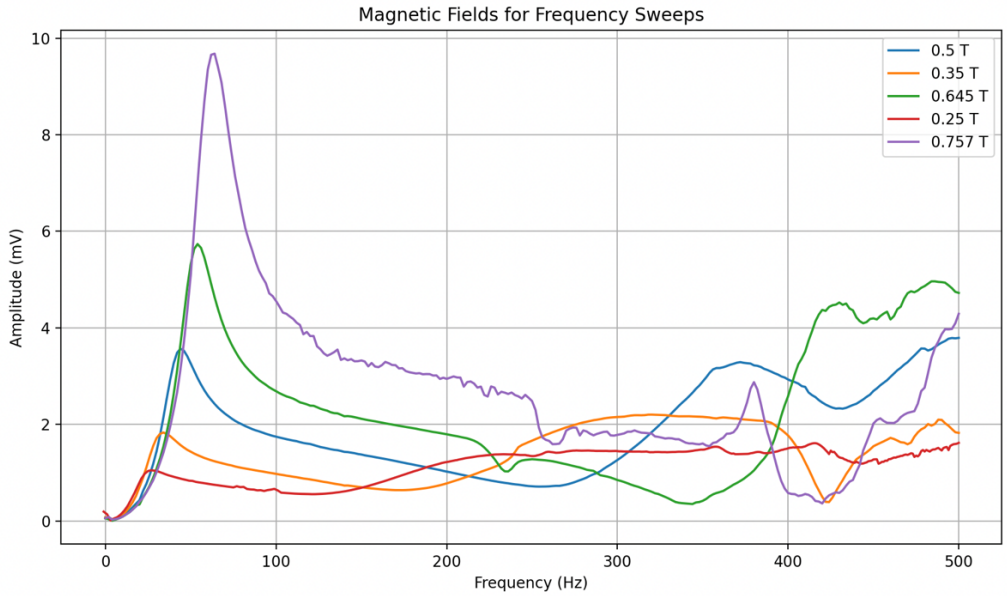
Date of submission: April 13, 2026
Instructors: Dr. Ken Clark, Dr. Jean-Michel Nunzi, Dr. Nahee Park

Melissa Choloniuk: 20005142

Executive Summary

The purpose of this experiment was to determine the Hall coefficient and resistivity of an indium sample that is used to generate Helicon waves. Helicon waves are low-frequency electromagnetic waves that are able to propagate through conducting medias, such as plasma, typically at low-temperatures [1], [2]. They are a sub-set of Whistler waves and were given the name "Helicon" for its spiral, helical shape [2]. The Helicon waves can be generated when a sample of pure indium metal is cooled to 4.2 K with a cryogenic liquid, such as liquid helium. An excitation coil placed parallel to this sample can excite Helicons with its oscillating magnetic field generated by the function generator [3].

Frequency sweeps were conducted for various magnetic field strengths (B_0= 0.25, 0.35, 0.5, 0.645, 0.757 T) and plotted. This can be seen in the figure below where a clear resonance peak for N=1 can be observed.



The data collected over the frequency sweeps was used to determine the resonant frequency and the full width at half-power. These values were ultimately used to calculate the Hall coefficient of 4.6 ± 1 x 10^{-10} m^3 C^{-1} for the sample of indium and an average resistivity of 5.9 ± 1 x 10^{-11} Ω·m. The expected accepted value for the Hall coefficient is 1.6 x 10^{-10} m^3 C^{-1}, giving an error of 9.8%, and resistivity is 1.5-1.7 x 10^{-11} Ω·m [4], [5]. Potential reasons for these discrepancies can be attributed to various sources of error, such as an unstable magnetic field strength reading and potentially impurities in the indium sample.

Table of Contents

Executive Summary ii

1 Data Analysis 1

2 Results and Discussion 4

3 Conclusion 5

Appendix 6

References 8

1 Data Analysis

This section interprets and analyzes the data collected over the course of the experiment. The frequency sweeps for the magnetic fields (B_0= 0.25, 0.35, 0.5, 0.645, 0.757 T) are plotted in Figure 1. A focus on the resonance peak N=1 can be seen in Figure 2 that shows the expected trend of having a stronger magnetic field strength yielding a bigger peak and higher frequency. The data in Figure 2 consists of a frequency range from 0 to 200 Hz to both highlight the N=1 resonance peak and reduce the following noise that can be seen in Figure 1.

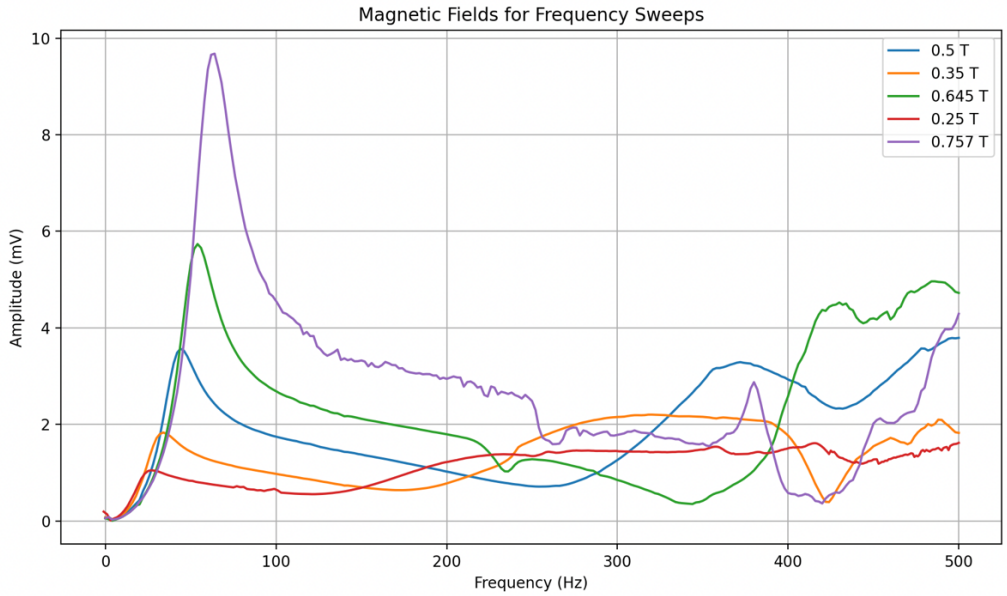


Figure 1: Frequency sweeps for magnetic field strengths B_0= 0.25, 0.35, 0.5, 0.645, 0.757 T.

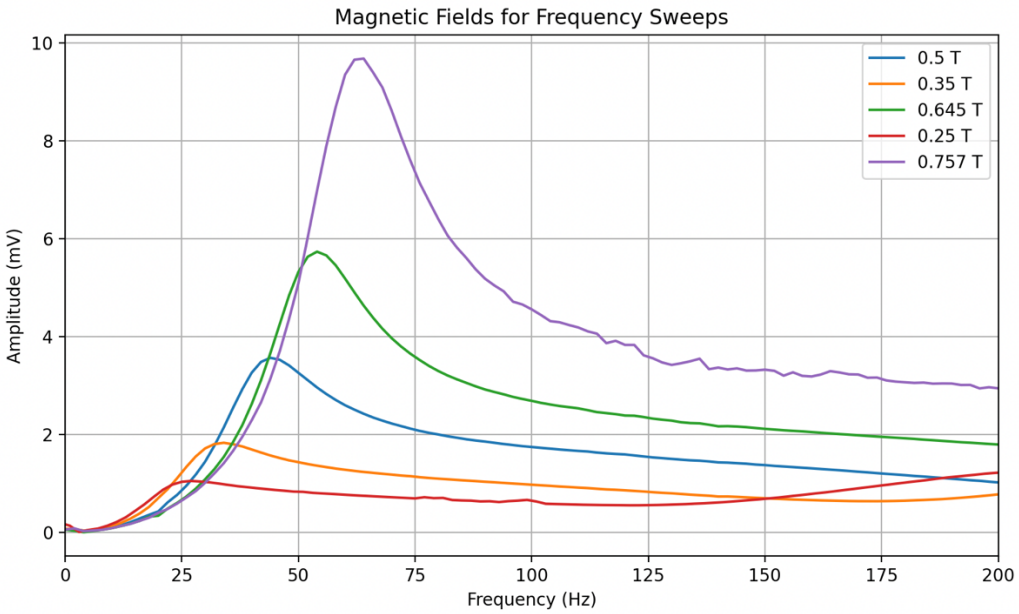


Figure 2: Frequency sweeps for magnetic field strengths B_0= 0.25, 0.35, 0.5, 0.645, 0.757 T focused on resonance peak N=1.

The resonant frequency, f_{res}, and the full width at half power, Δf, for each magnetic field are recorded in Table 1. The resonant frequency was found by finding the tangency point between the resonance curve and the line being fixed to the origin [4]. The full width half-power was found by taking the intersection of the resonance curve and a vertical line at $\frac{1}{\sqrt{2}}$ from the previously found resonance frequency [4]. This analysis was done for each magnetic field strength where an example for B_0=0.654 T can be seen in Figure 3 and the remaining plots can be found in the Appendix.

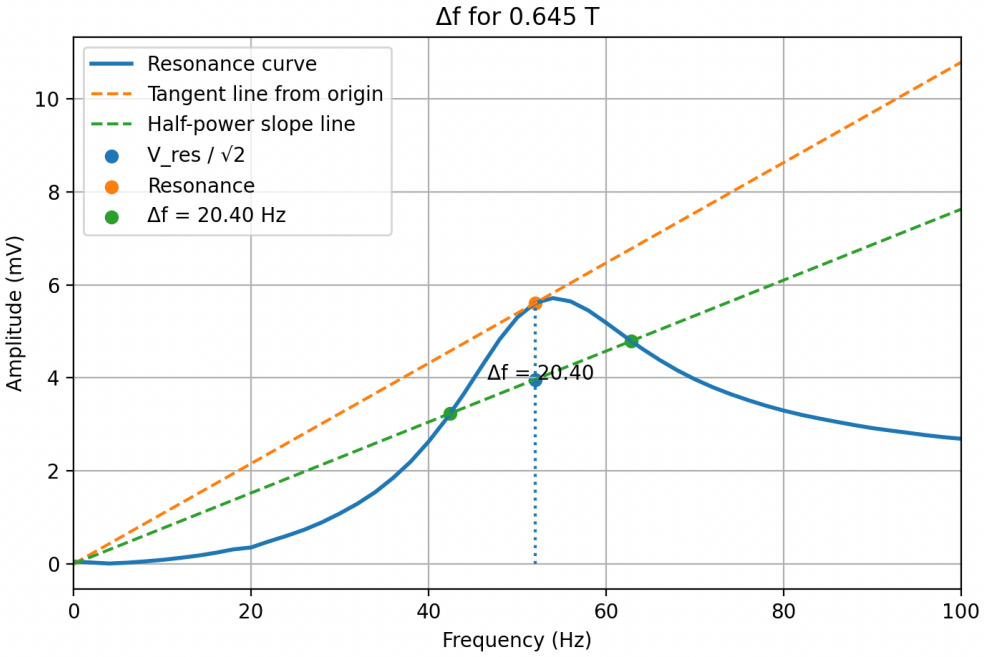


Figure 3: Frequency sweep for B_0=0.645 T focused on N=1 with the tangent and half-power slope line.

From this analysis the slope was taken from the line of best fit combined for each and found to be $76.3 \pm 2\ Hz/T$. The hall coefficient can be calculated using Equations 1 with the $\mu_0 = 4\pi x10^{-7} H/m$ and a thickness of the sample d=2.75mm [6].

$$R_H = \frac{2\mu_0 d^2}{N^2 \pi} \frac{df_{res}}{dB_0} \tag{1}$$

$$R_H = \frac{2(4\pi x10^{-7} H/m)(2.75x10^{-3} m)^2 (76.3 Hz/T)}{(1)^2 \pi}$$

$$R_H = 4.6x10^{-10} \pm 0.1\ m^3/C$$

The resistivity can be calculated for each magnetic field, B_0, using their respective resonance width at half-power, Δf, seen in Equations 2 [6]. A sample calculation for B_0=0.25 T can be seen below.

$$\rho = \frac{\mu_0 d^2 \Delta f}{N^2 \pi} \tag{2}$$

$$\rho = \frac{(4\pi x10^{-7} H/m)(2.75x10^{-3} m)^2 (17.9 \text{Hz})}{(1)^2 \pi}$$

$$\rho = 5.4x10^{-11} \pm 0.1 \Omega \cdot m$$

Table 1: Recorded values for the resonance frequency and resonance width at half-power for each magnetic field and the calculated values for each resistivity.

Magnetic Field, B_0 (T)	Resonant frequency, f_{res} (Hz)	Full width at half-power, Δf (Hz)	Resistivity, ρ ($\Omega \cdot m$)
0.25	22.9 ± 0.2	17.9 ± 0.7	$5.4 \times 10^{-11} \pm 0.1$
0.35	30.4 ± 0.1	19.2 ± 0.2	$5.8 \times 10^{-11} \pm 0.1$
0.5	41.8 ± 0.2	19.8 ± 0.2	$6.0 \times 10^{-11} \pm 0.1$
0.645	51.9 ± 0.2	20.4 ± 0.1	$6.2 \times 10^{-11} \pm 0.1$
0.757	61.0 ± 0.3	21.4 ± 0.1	$6.5 \times 10^{-11} \pm 0.1$

2 Results and Discussion

The experiment yielded an average Hall coefficient of $4.6 \pm 1 \times 10^{-10}$ $m^3\ C^{-1}$ and an average resistivity of $5.9 \pm 1 \times 10^{-11}$ $\Omega \cdot m$ for the sample of indium. The expected accepted value for the Hall coefficient is 1.6×10^{-10} $m^3\ C^{-1}$ and $\rho = 1.5\text{-}1.7 \times 10^{-11}$ $\Omega \cdot m$ [4], [5]. The experimental value obtained yielded an error of 9.8% for the Hall coefficient. This error could be attributed to many theorized factors. One such factor can be the reading of the magnetic field strength. As seen in Equation 1, the determination of the Hall coefficient is dependent upon the value of the magnetic field strength and so an incorrect reading of the magnetic field would directly impact the value of the Hall coefficient. The probe that is used to collect these readings can thus produce varying readings of B_0 depending on its location in the magnetic field and how close it is to the magnets. This was observed in the lab as, when the probe was moved, the Gaussmeter would read a wide range of magnetic field strengths.

Additional sources of error could also arise from inaccuracies with the indium sample. The sample is comprised of pure indium metal, however, of there were any impurities in the sample it could have an impact on the readings. Any impurities in the sample would impact its lattice resulting in its resistivity being increased [7]. In addition, if the dimensions of the sample were incorrect or have changed, this would result in an incorrect value for its thickness. As seen in Equation 1 when calculating the Hall coefficient, a change to the thickness of the sample would result in a different calculated value of the Hall coefficient. This is a possibility especially since the sample has to be cooled to 4.2K by means of a cryogenic liquid. Metals naturally shrink or contract under cold temperatures. The extent to which they shrink is dependent upon the coefficient of thermal expansion of the material to determine how much its dimensions change

[8]. Thus, the dimensions that were provided may not account for when it is cooled and thus the thickness may differ slightly.

3 Conclusion

The purpose of this experiment was to gain a better understanding of helicon waves and how they behave in metals, such as indium. This allowed for key fundamental aspects to be better investigated and observed, including the effects that the magnetic field strength directly has on the helicon resonance. Thus, from the generated magnetic field and performing frequency sweeps, the helicon resonance was able to be observed alongside the expected trend of the resonant frequency increasing with the strength of the magnetic field.

The Hall coefficient and resistivity of the indium were found from the frequency sweeps for each magnetic field (B_0= 0.25, 0.35, 0.5, 0.645, 0.757 T). The experimental value for the Hall coefficient was found to be 4.6 ± 1 x 10^{-10} m^3 C^{-1} for the sample of indium and an average resistivity of 5.9 ± 0.1 x 10^{-11} Ω·m. The Hall coefficient has a 9.8% error from the expected value of 1.6 x 10^{-10} m^3 C^{-1} for the Hall coefficient and the expected resistivity is 1.5-1.7 x 10^{-11} Ω·m.

There are various sources of error that could attribute to the error in the expected and experimental value. A primary source of error that would directly impact the Hall coefficient is the value of the magnetic field. The calculated value of the Hall coefficient is dependent upon the magnetic field, B_0, however as the probe located inside the magnetic field would shift, the reading on the Gaussmeter would change. Therefore, this causes shifts to the measuring of the resonant frequency and impacts the value of B_0. Additional errors could also come from any impurities in the sample of indium which would affect the Helicon waves and the value of its resistivity.

Appendix

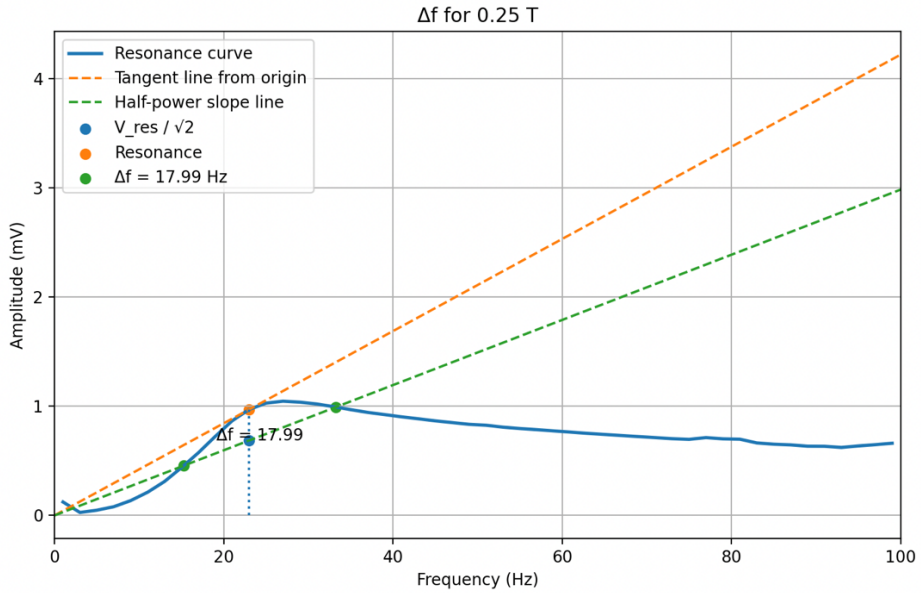


Figure 4: Frequency sweep for B_0=0.25 T focused on N=1 with the tangent and half-power slope line.

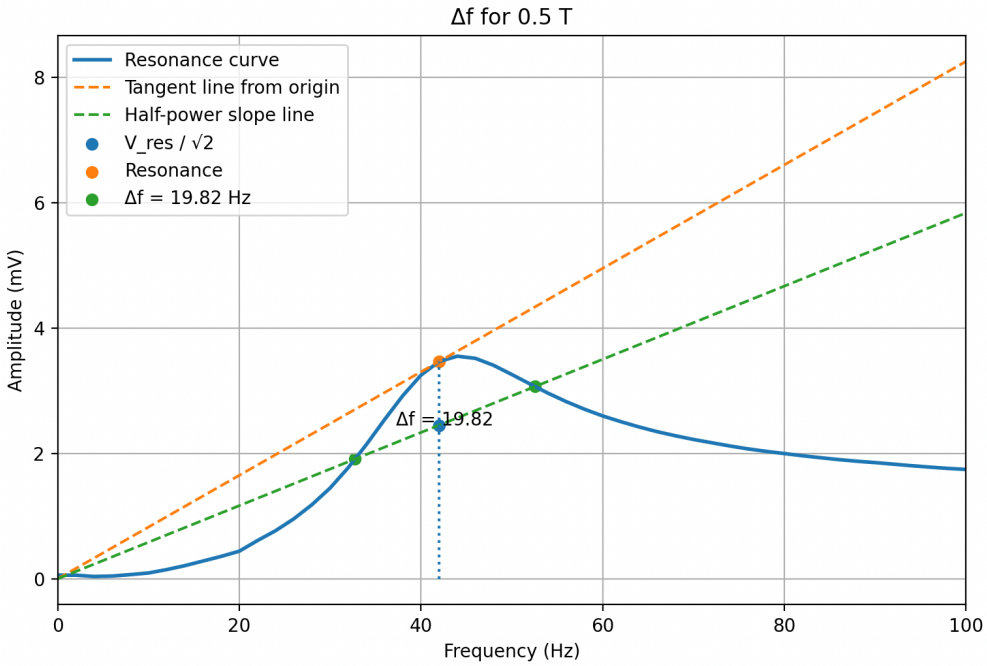


Figure 5: Frequency sweep for B_0=0.5 T focused on N=1 with the tangent and half-power slope line.

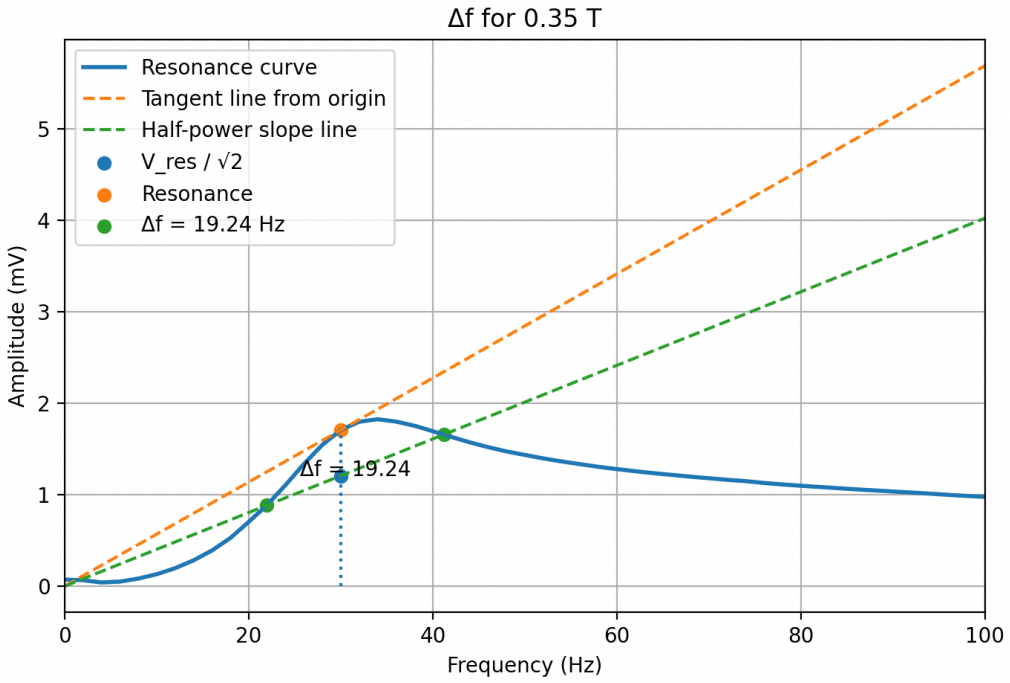


Figure 6: Frequency sweep for B_0=0.35 T focused on N=1 with the tangent and half-power slope line.

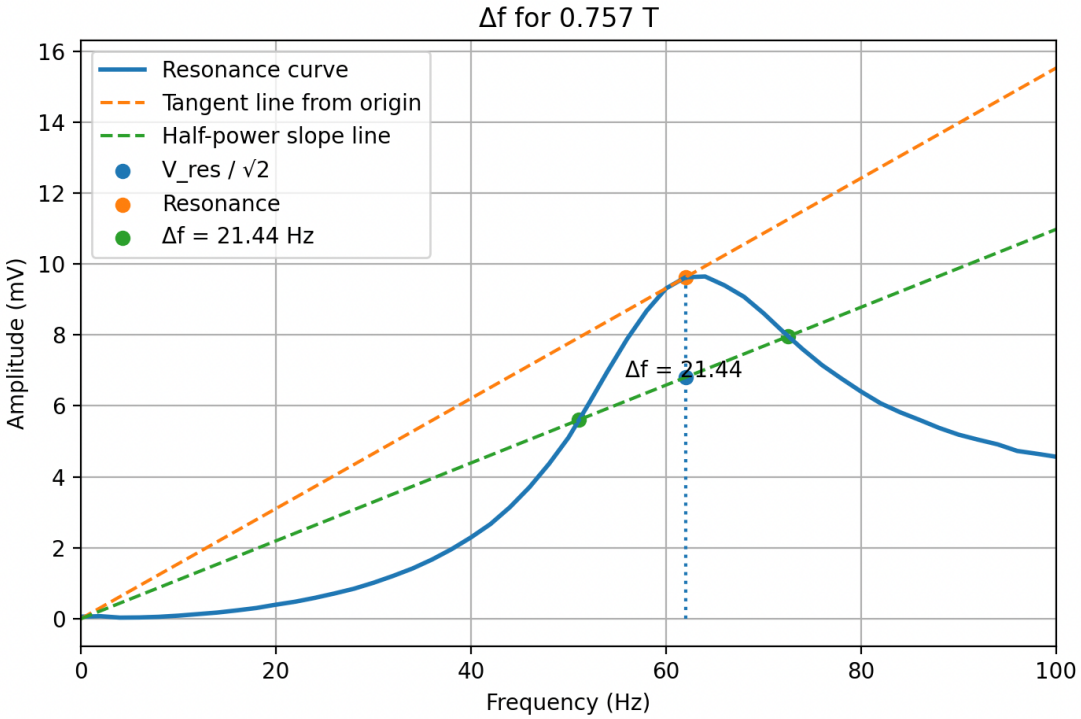


Figure 7: Frequency sweep for B_0=0.757 T focused on N=1 with the tangent and half-power slope line

References

[1] N. Stumpf, “INITIAL DEVELOPMENT OF A HELICON PLASMA SOURCE,” Dec. 2021, [Online]. Available: https://brightspotcdn.byui.edu/6d/57/b02124c9451892531b4562bf8379/nathaniel-stumpff21.pdf

[2] R. L. Stenzel, “Whistler waves with angular momentum in space and laboratory plasmas and their counterparts in free space,” *Adv. Phys. X*, vol. 1, no. 4, pp. 687–710, Jul. 2016, doi: 10.1080/23746149.2016.1240017.

[3] L. Tang, Q. Zhang, Q. Lu, X. Gao, and Y. Ke, “Experimental Study on the Properties of Helicon Mode Whistler Waves,” *Chin. J. Space Sci.*, vol. 44, pp. 1006–1011, Nov. 2024, doi: 10.11728/cjss2024.06.2024-yg30.

[4] J. R. Merrill, D. Pierce, and D. Giovanielli, “A Helicon Solid-State Plasma Experiment for the Advanced Laboratory,” *Am. J. Phys.*, vol. 38, no. 4, pp. 417–421, Apr. 1970, doi: 10.1119/1.1976357.

[5] J. M. Goodman, “Helicon Waves, Surface-Mode Loss, and the Accurate Determination of the Hall Coefficients of Aluminum, Indium, Sodium, and Potassium,” *Phys. Rev.*, vol. 171, no. 3, pp. 641–658, Jul. 1968, doi: 10.1103/PhysRev.171.641.

[6] John Pitre, “Helicons in Metals,” 2012, [Online]. Available: https://www.physics.utoronto.ca/apl/hel/hel.pdf

[7] “Effet des impuretes sur la resistivite electrique du plutonium en phase alpha a basses temperatures: — Correlation entre la purete chimique et la resistivite residuelle — Cas particulier de l’americium - ScienceDirect.” Accessed: Apr. 13, 2026. [Online]. Available: https://www-sciencedirect-com.proxy.queensu.ca/science/article/pii/0022311576901677?via%3Dihub

[8] “Why Does Metal Shrink When Cold? The Science Explained,” Biology Insights. Accessed: Apr. 13, 2026. [Online]. Available: https://biologyinsights.com/why-does-metal-shrink-when-cold-the-science-explained/